

Mid 1

Question 1:

1. Identify the Performance Issue

The key performance issue is **high latency for international users** due to the centralized hosting model:

- **All content (static and dynamic)** is being served from a **single US-based data center**.
- **Users in Asia, Europe, and South America** experience **longer round-trip times** for requests.
- **Static assets** (images, CSS, videos) take longer to load due to the physical distance and possible congestion.

2. Vertical scaling/horizontal scaling/Use a Content Delivery Network (CDN)

3. **Best initial solution: Use a Content Delivery Network (CDN)**

Question 2:

<body>

<h2>Applicant Search</h2>

<input type="text" id="applicantId" placeholder="Enter applicant ID" />

<button onclick="searchApplicant()">Search</button>

<div id="applicantContainer"></div>

<script>

function searchApplicant() {

const applicantId =
document.getElementById("applicantId").value.trim();

const container =
document.getElementById("applicantContainer");

if (!applicantId) {

```

        container.textContent = "Please enter an applicant ID.";

        container.className = "error";

        return;
    }

    container.textContent = "Loading...";

    container.className = "";

    fetch(`https://backend.hireTalent.io/applicants/search?applicantId=${
        encodeURIComponent(applicantId)}`)

        .then(response => response.json())

        .then(data => {

            if (data.name) {

                container.innerHTML = `

                    <strong>Name:</strong> ${data.name}<br/>

                    <strong>Email:</strong> ${data.email}<br/>

                    <strong>Experience:</strong> ${data.experience}

                `;

                container.className = "";

            } else {

                container.textContent = data.message || "Invalid
applicant ID";

                container.className = "error";

            }

        })

        .catch(error => {

            container.textContent = "An error occurred. Please try
again.";

```

```
        container.className = "error";

    });

}

</script>

</body>
```

Question 3

Start

Inside async function

Call at the end

End

Promise 1

Promise 2

After await

Timeout 2

Timeout 1